

E-A.I Platform — Backend Specification

Version 1.0 **Date** June 2026 **Product** Etech / E-A.I **Scope** Backend API & Infrastructure

1. OVERVIEW

E-A.I is an AI-powered course generation platform. Users describe a topic, choose a format (text, audio, image, interactive, video) and a difficulty level, and the platform generates a complete structured course including lessons, quizzes, diagrams, exercises, and a capstone project.

The backend is responsible for user authentication, course generation orchestration via an async job pipeline, AI service integration, user progress tracking, and file storage for generated media assets.

2. USER FLOWS → BACKEND MAPPING

Frontend Screen	Route	Backend Requirement
Sign Up form	/sign-up	POST /auth/register
Login form → /dashboard	/login	POST /auth/login → JWT issued
Forgot Password	/login	POST /auth/forgot-password → email sent
Dashboard: pick format	/dashboard	Format stored on generation session
Generation: type prompt	/generate	Prompt passed to generation API
Topic + Level + progress bar	/select-topic	POST /courses/generate → async job
Results page (not yet built)	/results	GET /courses/:courseId
Profile page (empty shell)	/profile	GET /user/profile, GET /user/courses, GET /user/progress

3. AUTHENTICATION AUTH

Endpoints

```
POST    /api/v1/auth/register
POST    /api/v1/auth/login
POST    /api/v1/auth/logout
POST    /api/v1/auth/forgot-password
POST    /api/v1/auth/reset-password
POST    /api/v1/auth/refresh-token
```

Register Payload

```
{
  "firstName": "string",
  "lastName": "string",
  "email":     "string",    // unique, validated by regex
  "phone":    "string",    // optional
  "password": "string"     // min 8 chars
}
```

Login Payload & Response

```
// Request
{ "email": "string", "password": "string" }

// Response
{
  "accessToken": "JWT",          // 15-minute TTL
  "refreshToken": "string",      // 7-day TTL, rotated on use
  "user": {
    "id": "uuid",
    "firstName": "string",
    "lastName": "string",
    "email": "string",
    "phone": "string"
  }
}
```

Security Requirements

- Passwords hashed with bcrypt (minimum cost factor 12)
- Email uniqueness enforced at DB level
- Email validation: `/^[^\s@]+@[^\s@]+\.[^\s@]+$/`
- JWT access token short-lived (15m); refresh token rotated on every use
- Forgot-password flow: time-limited one-use token sent via email (expire 1hr)
- All auth-protected routes require `Authorization: Bearer <token>`

4. COURSE GENERATION ASYNC AI

Course generation is the core feature. It is async — the frontend polls progress via the job status endpoint while displaying a 0-100% progress bar.

Endpoints

POST	/api/v1/courses/generate	→ kick off job, return jobId
GET	/api/v1/courses/generate/:jobId	→ poll progress
GET	/api/v1/courses/:courseId	→ fetch completed course
GET	/api/v1/courses	→ list user's courses (paginated)
DELETE	/api/v1/courses/:courseId	→ delete a course

Generate Request

```
{
  "prompt": "string",           // user's free-text prompt
  "topic": "string",           // e.g. "Python Basics"
  "level": "beginner | intermediate | advanced",
  "format": "text | audio | image | interactive | video",
  "duration": "30-45mins | 1-2 Hours | Full Course"
}
```

Generate Response (immediate)

```
{
  "jobId": "uuid",
  "status": "queued",
  "estimatedSeconds": 30
}
```

Progress Poll Response

```
{
  "jobId": "uuid",
  "status": "queued | processing | complete | failed",
  "progress": 0-100,
  "courseId": "uuid | null", // populated when status = complete
  "error": "string | null" // populated when status = failed
}
```

Completed Course Structure

```
{
  "id": "uuid",
  "title": "string",
  "topic": "string",
  "level": "beginner",
  "format": "text",
  "duration": "1-2 Hours",
  "lessons": [
    {
```

```

    "id": "uuid",
    "order": 1,
    "title": "string",
    "content": "string | html",
    "diagrams": [{ "url": "string", "caption": "string" }],
    "quiz": {
      "questions": [
        { "q": "string", "options": [], "answer": "string" }
      ]
    },
    "exercises": [{ "title": "string", "description": "string" }]
  },
  "project": {
    "title": "string",
    "description": "string",
    "requirements": ["string"]
  },
  "audioUrl": "string | null", // for audio format
  "createdAt": "ISO8601"
}

```

AI Pipeline — Per Format

Format	Pipeline
text	LLM → structured lessons + quizzes + exercises + project
audio	LLM generates content → TTS converts each lesson to audio file → uploaded to object storage
image	LLM generates content → diagram/image generation API produces visuals per lesson
interactive	LLM generates structured step-through Q&A / dialogue format per lesson
video	LLM generates script → video synthesis service (or store script for manual production)

Rate Limiting

- Max 5 concurrent generation jobs per user
- Generation endpoint: 10 requests/minute per user

5. USER PROFILE & PROGRESS

Endpoints

GET	/api/v1/user/profile	
PUT	/api/v1/user/profile	
GET	/api/v1/user/courses	→ paginated course history
GET	/api/v1/user/progress	→ aggregate learning stats
GET	/api/v1/user/progress/:courseId	→ per-course progress
PUT	/api/v1/user/progress/:courseId	→ mark lesson / quiz complete

Profile Update Payload

```
{
  "firstName": "string",
  "lastName": "string",
  "phone": "string",
  "avatarUrl": "string" // URL after upload to object storage
}
```

Progress Update Payload

```
{
  "lessonId": "uuid",
  "completed": true,
  "quizScore": 0-100 // optional
}
```

Progress Response

```
{
  "courseId": "uuid",
  "overallPercent": 72,
  "lessonsCompleted": ["uuid", "uuid"],
  "quizScores": { "lessonId": 85 },
  "projectCompleted": false,
  "lastAccessedAt": "ISO8601"
}
```

6. DATA MODELS

User

Field	Type	Notes
id	UUID	Primary key
firstName	string	
lastName	string	
email	string	Unique, indexed
phone	string	Nullable
passwordHash	string	bcrypt
avatarUrl	string	Nullable
createdAt	timestamp	
updatedAt	timestamp	

Course

Field	Type	Notes
id	UUID	Primary key
userId	UUID	FK → User
prompt	text	Original user prompt
topic	string	
level	enum	beginner / intermediate / advanced
format	enum	text / audio / image / interactive / video
duration	enum	30-45mins / 1-2 Hours / Full Course
status	enum	queued / processing / complete / failed
content	jsonb	Full course structure
createdAt	timestamp	
completedAt	timestamp	Nullable

Progress

Field	Type	Notes
id	UUID	Primary key
userId	UUID	FK → User
courseId	UUID	FK → Course

lessonsCompleted	uuid[]	Array of completed lesson IDs
quizScores	jsonb	<code>{ lessonId: score }</code>
projectCompleted	boolean	Default false
overallPercent	int	Computed on update
lastAccessedAt	timestamp	

GenerationJob

Field	Type	Notes
id	UUID	Primary key / jobId
userId	UUID	FK → User
courseId	UUID	FK → Course, set on complete
status	enum	queued / processing / complete / failed
progress	int	0-100
error	text	Nullable
createdAt	timestamp	
updatedAt	timestamp	

7. API DESIGN STANDARDS

- **Base URL:** `/api/v1`
- **Auth:** Bearer JWT on all routes except `/auth/*` and public landing data
- **Pagination:** Cursor-based — `?cursor=&limit=`
- **Content-Type:** `application/json` on all endpoints

Standard Error Shape

```
{
  "error": "Human-readable message",
  "code": "ENUM_CODE",           // e.g. INVALID_EMAIL, JOB_LIMIT_EXCEEDED
  "details": {}                  // optional field-level errors
}
```

HTTP Status Codes

Code	When
200	Successful GET / PUT
201	Resource created (register, course started)
400	Validation error
401	Missing or invalid token
403	Accessing another user's resource
404	Resource not found
429	Rate limit exceeded
500	Server / AI service error

8. EXTERNAL SERVICES

Service	Purpose	Suggested Provider
LLM API	Course content, quizzes, exercises, projects	Claude (Anthropic) / OpenAI GPT-4o
Text-to-Speech	Audio format — convert lessons to audio files	ElevenLabs / AWS Polly
Diagram Generation	Diagrams & visuals feature	Mermaid (rendered server-side) / DALL·E
Object Storage	Audio files, generated images, avatars	AWS S3 / Cloudflare R2
Job Queue	Async generation pipeline	BullMQ + Redis
Email Service	Password reset, welcome emails	Resend / SendGrid
Database	Primary data store	PostgreSQL
Cache	Job state, rate limiting, sessions	Redis

9. OUT OF SCOPE (V1) — FUTURE HOOKS

Feature	Frontend Signal
/results route	Referenced in TopicSelectionView but page not yet built
Social auth (Google/GitHub)	Not in UI yet — common for this product type
Course sharing / public links	Not referenced yet
Payments / subscription tiers	Would gate generation limits and formats
Video generation pipeline	Video format selected on dashboard but pipeline TBD